Project Planck Submission — Paralog Redundancy / Curative Threshold

Prompt (submit this block)

In a panel of six cultured tumor cell lines (L1–L6) from a hypothetical carcinoma, three paralogous kinase proteins, K1, K2, and K3, are co-expressed. siK1, siK2, and siK3 are validated siRNAs; each silences expression of only its correspondingly numbered protein, confirmed by immunoblot to reduce that protein to under 10% of control, with equivalent knockdown efficacy across all three and no detectable reduction of either other paralog.

Treatment with siK1 alone, siK2 alone, siK3 alone, siK1+siK2, or siK2+siK3 each left proliferation and a cell-death marker statistically indistinguishable from non-targeting control (death-marker below 0.20, normalized to a cytotoxic positive control set at 1.00) in all six lines.

Treatment with siK1+siK3 gave, per line, proliferation (% of control) and death-marker: L1 98%/0.12, L2 45%/0.22, L3 52%/0.18, L4 95%/0.10, L5 101%/0.15, L6 40%/0.25.

Treatment with siK1+siK2+siK3 gave, per line: L1 8%/0.82, L2 5%/0.91, L3 6%/0.88, L4 9%/0.79, L5 7%/0.85, L6 4%/0.93.

Define a curative response in a given line as death-marker exceeding 0.60 in that line. Proliferation reduction alone, without exceeding this threshold, indicates growth suppression, not cell killing.

Using only the data above, in how many of the six cell lines does the one protein left unsilenced by the siK1+siK3 treatment fail to prevent a curative response? Report as a single integer.

Character count: 1,404 / 2,000 limit

Answer Format

Integer, 0–6.

Final Answer

0

Golden Response

Step 1: Identify which protein is unsilenced by the siK1+siK3 treatment.
The prompt states there are exactly three paralogous proteins, K1, K2, K3. siK1 silences K1; siK3 silences K3; neither siRNA reduces the other paralogs (stated explicitly). Silencing K1 and K3 together therefore leaves K2 as the only one of the three proteins not silenced. This identification is not stated directly in the prompt — it must be derived from the set of three proteins minus the two named as targeted.

Step 2: Identify what "curative" means as defined in the prompt.
A curative response in a given cell line is defined as death-marker exceeding 0.60 in that line. Proliferation percentage is a separate measurement and is explicitly stated not to indicate cell killing on its own, regardless of how low it drops.

Step 3: Isolate the correct data series.
The question concerns the siK1+siK3 treatment (which leaves K2 unsilenced, per Step 1). The relevant six data points are the death-marker values from that treatment only: L1 0.12, L2 0.22, L3 0.18, L4 0.10, L5 0.15, L6 0.25.

Step 4: Apply the stated threshold to each line.
L1: 0.12 < 0.60 — does not meet threshold
L2: 0.22 < 0.60 — does not meet threshold

L3: 0.18 < 0.60 — does not meet threshold
L4: 0.10 < 0.60 — does not meet threshold
L5: 0.15 < 0.60 — does not meet threshold
L6: 0.25 < 0.60 — does not meet threshold

None of the six lines cross the death-marker threshold under siK1+siK3, despite proliferation being substantially reduced in three of them (L2, L3, L6). A response that only reasons from the proliferation numbers, without first correctly identifying that K2 is the protein left unsilenced and without applying the stated death threshold, risks answering 3 — treating growth suppression in three lines as if it were a curative response in those same three lines. This is the reasoning error the question is designed to catch.

Step 5: Final synthesis.
Leaving K2 as the sole unsilenced protein fails to prevent a curative response in zero of the six lines — because siK1+siK3 never produces a curative (death-threshold-crossing) response in any line to begin with, so there is no line where K2's presence "fails to prevent" one. Zero of the six lines show K2 alone falling short of blocking a curative outcome, since no curative outcome occurs under this treatment in any line for K2 to have failed to block.

Final Answer: 0

Notes on Originality / Design (not part of submission)

- K1/K2/K3, siK1/siK2/siK3, and the six-line panel are entirely fictional; no resemblance to any specific real gene family, paper, or dataset intended beyond the general logical structure (paralog redundancy, cell-line-dependent sufficiency, growth-suppression-vs-death distinction).
- Verify exact character count of the Prompt block before submission — trim if over 2000.
- Two reasoning steps are deliberately required, not one: (1) derive which protein is left unsilenced by a named siRNA combination (never stated directly), and (2) apply the death-marker threshold rather than the proliferation numbers to judge "curative." A shallow response that skips step 1 or conflates growth suppression with death in step 2 will fail — this maps to the platform's "flawed causal reasoning" and "missing critical steps" failure categories.
- The final-question wording ("fails to prevent a curative response") needs one more pass — check it doesn't read as circular or ambiguous before submission; consider simplifying to directly ask "in how many of the six lines does the siK1+siK3 treatment cross the curative threshold" instead, which tests the same two reasoning steps with a cleaner, less circular question.